The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)

University College London Hospitals NHS Foundation Trust

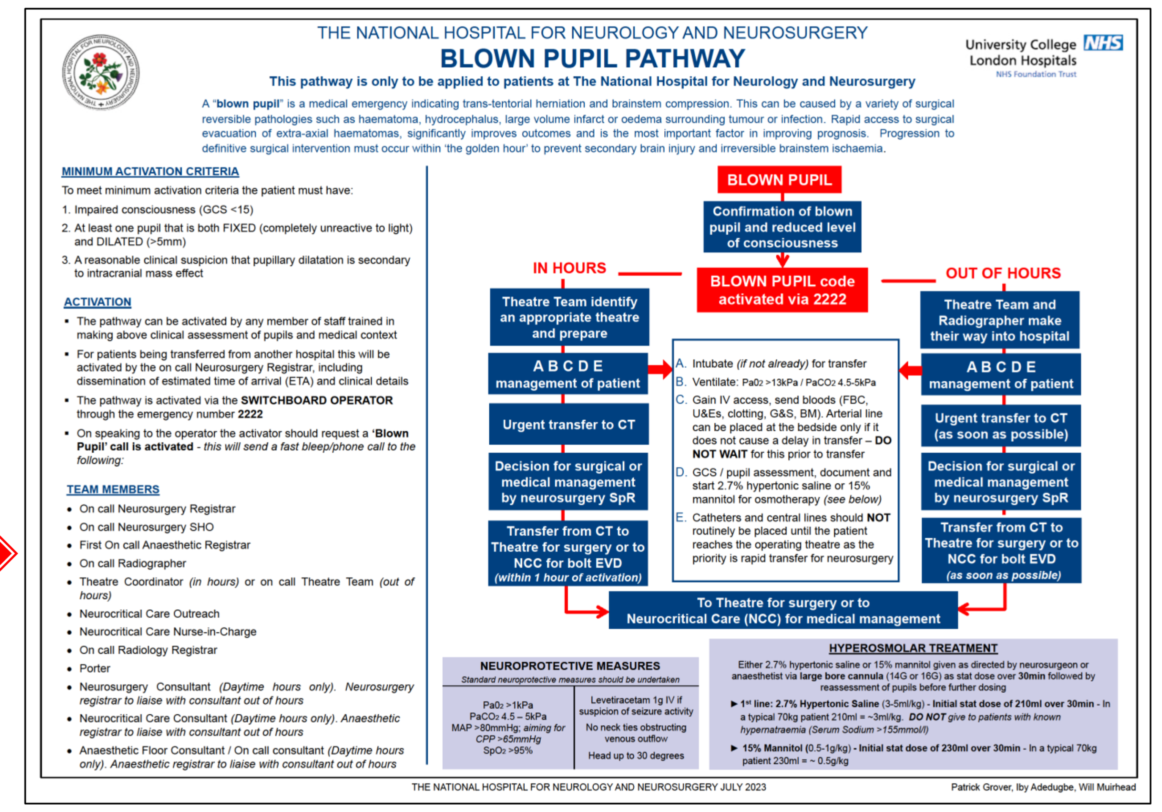
HYPEROSMOLAR THERAPY

HYPERTONIC SALINE 2.7% and MANNITOL 15%

Both 2.7% hypertonic saline (HTS) and mannitol 15% can effectively reduce intracranial pressure, but HTS has a more sustained effect on intracranial pressure and can effectively increase cerebral perfusion pressure and is therefore the 1st choice hyperosmolar therapy.

Hyperosmolar therapy should be given as directed by neurosurgeon, NCCU consultant or anaesthetist via **large bore cannula** (14G or 16G) as stat dose over **30min** followed by reassessment of pupils before further dosing.

Hyperosmolar therapy is most commonly used in the event of a 'blown pupil'. A "**blown pupil**" is a medical emergency indicating trans-tentorial herniation and brainstem compression. This can be caused by a variety of surgical reversible pathologies such as haematoma, hydrocephalus, large volume infarct or oedema surrounding tumour or infection. Rapid access to surgical evacuation of extra-axial haematomas, significantly improves outcomes and is the most important factor in improving prognosis. Progression to definitive surgical intervention must occur within 'the golden hour' to prevent secondary brain injury and irreversible brainstem ischaemia. In event of a 'blown pupil' follow BLOWN PUPIL PATHWAY



THE NATIONAL HOSPITAL FOR NEUROLOGY AND NEUROSURGERY
BLOWN PUPIL PATHWAY
This pathway is only to be applied to patients at The National Hospital for Neurology and Neurosurgery

A "**blown pupil**" is a medical emergency indicating trans-tentorial herniation and brainstem compression. This can be caused by a variety of surgical reversible pathologies such as haematoma, hydrocephalus, large volume infarct or oedema surrounding tumour or infection. Rapid access to surgical evacuation of extra-axial haematomas, significantly improves outcomes and is the most important factor in improving prognosis. Progression to definitive surgical intervention must occur within 'the golden hour' to prevent secondary brain injury and irreversible brainstem ischaemia.

MINIMUM ACTIVATION CRITERIA
To meet minimum activation criteria the patient must have:
1. Impaired consciousness (GCS <15)
2. At least one pupil that is both FIXED (completely unreactive to light) and DILATED (>5mm)
3. A reasonable clinical suspicion that pupillary dilatation is secondary to intracranial mass effect

ACTIVATION
- The pathway can be activated by any member of staff trained in making above clinical assessment of pupils and medical context
- For patients being transferred from another hospital this will be activated by the on call Neurosurgery Registrar, including

BLOWN PUPIL
Confirmation of blown pupil and reduced level of consciousness
BLOWN PUPIL code activated via 2222
IN HOURS
OUT OF HOURS
Theatre Team identify an appropriate theatre and prepare
Theatre Team and Radiographer make their way into hospital
A B C D E
A. Intubate (if not already) for transfer

1st LINE: 2.7% HYPERTONIC SALINE (3-5ml/kg)

***DO NOT** give to patients with known hypernatraemia (Serum Sodium >155mmol/L)*

Effects and Evidence for use

- Hypertonic saline (HTS) restores intravascular volume and may increase BP as well as lowering ICP
- HTS has a rapid onset, ≈ 5 minutes, with effects lasting up to 12 hours without rebound ICP increase (may be some rebound increase in ICP but this side effect is limited)
- HTS provides a more rapid and sustained ICP reduction compared with mannitol
- Osmotic effect - mechanism of action predominantly through marked osmotic shift of fluid from intracellular to interstitial and intravascular space
- Hypertonic saline causes hypernatraemia – therefore serum sodium should be measured within 6 hours of HTS administration if bolus doses given and measured regularly if infusion commenced

STEP 1

Initial stat dose of 210ml over 30min

- In a typical 70kg patient 210ml = ~3ml/kg

STEP 2

Reassessment of pupils prior to further dosing on medical instruction

- Re-administration of hypertonic saline should not occur until serum sodium concentration is <155mmol/L
- Acute hyperosmolar state - most feared complication is central pontine myelinolysis, also known as ODS (osmotic demyelination syndrome) in hyponatremic states when sodium levels have been corrected rapidly
- Extravasation injuries may cause marked local tissue injury, therefore check infusion site regularly or give via central line if available

15% MANNITOL (0.5-1g/kg)

Sugar alcohol hypertonic solution

Effects and Evidence for use

- Reduces blood viscosity, increases CBF and increases cerebral oxygen delivery
- Reduces ICP within a few minutes *(most marked when CPP low (<70mmHg)*
- Osmotic diuretic (*effect delayed for 15-30min, persists for 90min to 6h or more)* - can cause hypotension secondary to osmotic diuresis, which could potentially be deleterious in patients who are hypotensive or hypovolaemic
- Large doses of Mannitol *1.8g/kg-2.1g/kg* effective in comatose patients with operative haematoma *(Level I evidence = High degree of clinical certainty)*
- 1.2-1.4g/kg in comatose trauma patients with operative ASDH or operative intraparenchymal temporal lobe haemorrhage *(Level I recommendation)*
- 0.25-1g/kg to control elevated ICP in severe TBI *(Level II recommendation)* infused over at least 15 minutes *(May be most effective when CPP <70mmHg (Level II recommendation)*
- Bolus dose at rate not exceeding 0.1g/kg/min to avoid risk of sudden hypotension (*Level II recommendation)*
- Intermittent boluses may be more effective than continuous infusion (*Level III recommendation)*

STEP 1

Initial stat dose of 230ml over 30min

- In a typical 70kg patient 230ml = ~ 0.5g/kg

STEP 2

Reassessment of pupils prior to further dosing on medical instruction

Mannitol intoxication and the 'Rebound Effect'

- When infusion rate exceeds rate of hourly excretion in urine, mannitol accumulates in extracellular space
- Breakdown of blood-brain barrier and extravasation into interstitium of brain thought to result in reverse osmotic gradient that can cause cerebral oedema
- Risk may be greater with continuous rather than bolus infusions
- Monitor electrolytes, fluid balance and serum and urine osmolarity

Calculations for bolus dose based on lowest recommended doses

SODIUM CHLORIDE 2.7% BOLUS DOSE	
WEIGHT (kg)	ml for 3ml/kg
40	120
45	135
50	150
55	165
60	180
65	195
70	210
75	225
80	240
85	255
90	270
95	285
100	300
110	330
>120	360

MANNITOL 15% BOLUS DOSE	
WEIGHT (kg)	ml for 0.5g/kg
40	130
45	150
50	165
55	180
60	200
65	215
70	230
75	250
80	265
85	280
90	300
95	315
100	330
110	365
>120	400

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